

EXPERIMENT 20

Implement a basic information retrieval system using TF-IDF (Term Frequency-Inverse Document Frequency) for document ranking using python

PROGRAM:

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "Machine learning is interesting",
    "Artificial intelligence and machine learning",
    "Python programming language",
    "Data science uses machine learning"
]

query = ["machine learning"]
vectorizer = TfidfVectorizer()
tfidf = vectorizer.fit_transform(documents + query)
similarity = cosine_similarity(tfidf[-1], tfidf[:-1])
scores = similarity.flatten()
print("Document Ranking:")
ranking = sorted(enumerate(scores), key=lambda x: x[1], reverse=True)
for index, score in ranking:
    print(f"Document {index+1}: Score = {score:.3f}")
```

OUTPUT:

```
print("Document Ranking:")

ranking = sorted(enumerate(scores), key=lambda x: x[1], reverse=True)

for index, score in ranking:
    print(f"Document {index+1}: Score = {score:.3f}")
```

... Document Ranking:
Document 1: Score = 0.491
Document 2: Score = 0.418
Document 4: Score = 0.418
Document 3: Score = 0.000